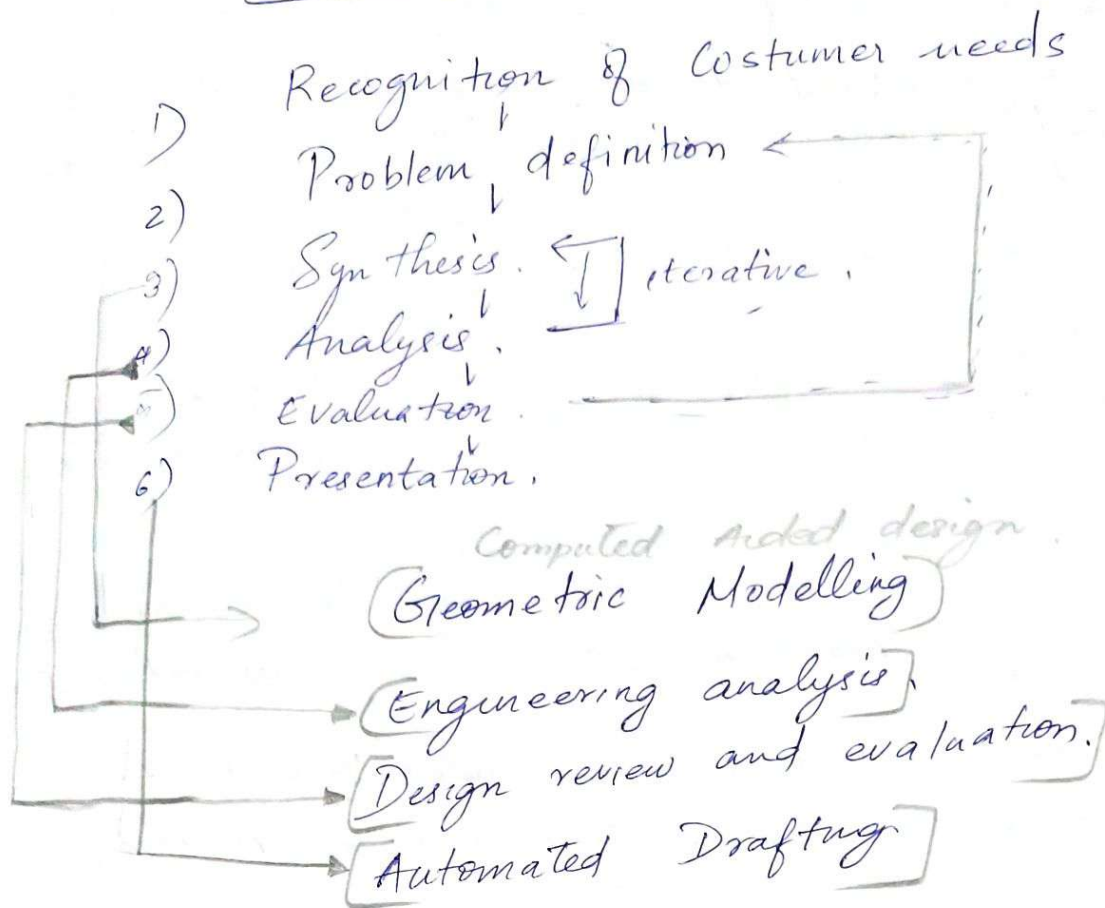


Product Development



Design process consist of 6 steps.

- ① Recognition of need : Involves the realisation by some one that a problem exist for which some collective action should be taken. This might be the identification of some defects in the current design.
- ② Definition of a problem : Involves the specification of the item to be designed, includes physical and functional characteristics, cost quality and operating performance.
- ③ Synthesis : A certain component or subsystem of the overall system is conceptualised by the design.

④ Analysis and optimisation: Synthesis and analysis are closely related & highly iterative.

The conceptualised design subjected to analysis, improved through analysis procedure and re designed, The process is repeated until the design has been optimised.

⑤ Evaluation, Measuring the design against the specification established in the problem definition faced.

⑥ Presentation: Documentation of design by means of drawings, material specifications, assembly list etc.

In General: CAD (Computer aided design). Implementation of computer application into various process.

The geometrical modelling corresponds to synthesis phase.
Engineering analysis corresponds to analysis and optimisation,

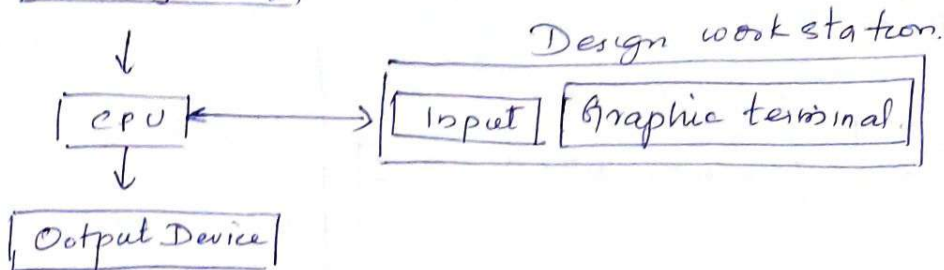
Design review and evaluation corresponds to 5th phase and Automated drafting involves a procedure for converting Design Image data in the computer memory into hard copy document.

10/1/24. Screen we see is GUI - Graphical, user Interface.

CAD: Computer Aided design.

Involves any type of design activity in which that make use of the computer to develop, analyse, or modify engineering design. The modern CAD system. Based on interactive Computer Graphics (ICG)

Secondary storage



ICG denotes a user oriented system in which the computer is employed to create transform and display in the form of pictures and symbols. The image is constructed out of basic geometric elements (point, line, circle). It can be modified according to the command of the designer like enlarge, reduce in size, move to another location, rotate, transform.

The user in the ICG who communicates data and commands to the computer through several input devices called designer.

A typical ICG system is a combination of hardware and software.

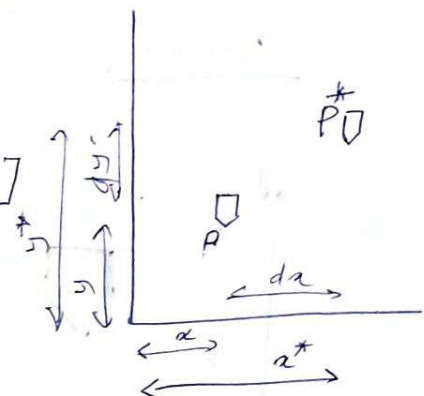
Translation

Initial position $P = [x, y]$

final position $P^* = [x^*, y^*]$

$$[P^*] = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} x + dx \\ y + dy \end{bmatrix}$$

$$[P^*] = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} dx \\ dy \end{bmatrix}$$



Geometric transformation

$$\Rightarrow [P^*] = [P] [T_t]$$

T_t = Transformation matrix for translation

Geometric transformation Convert geometry from one coordinate system to another.

Rotation

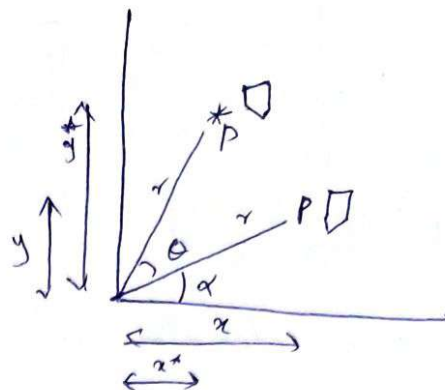
$$P^* = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} r \cos(\alpha + \theta) \\ r \sin(\alpha + \theta) \end{bmatrix}$$

$$= \begin{bmatrix} r \cos \theta - y \sin \theta \\ r \sin \theta + x \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} \cos \theta - \sin \theta \\ \sin \theta \cos \theta \end{bmatrix}$$

$$[P^*] = [P] [T_r]$$

for rotation in clock wise θ is negative
for anti-clockwise direction θ is considered +ve.



$$x = r \cos \alpha$$

$$y = r \sin \alpha$$

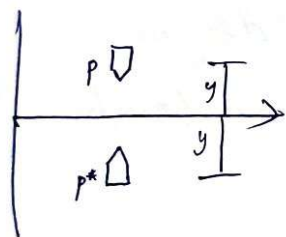
$$x^* = r \cos(\theta + \alpha)$$

$$y^* = r \sin(\theta + \alpha)$$

$$\Rightarrow x^* = r \cos \alpha \cos \theta - r \sin \alpha \sin \theta$$

$$y^* = r \sin \alpha \cos \theta + r \cos \alpha \sin \theta$$

Reflection / mirror



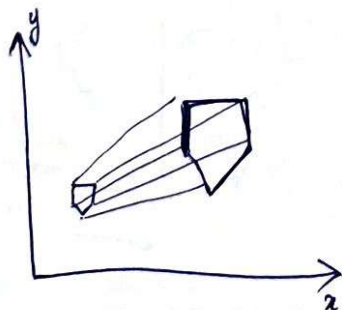
$$P^* = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$[P^*] = [T_m] [P]$$

$$x^* = x \cos \theta - y \sin \theta$$

$$y^* = x \sin \theta + y \cos \theta$$

Scaling



$$P^* = [x^*, y^*] = [s_x x, s_y y]$$

$$[P^*] = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

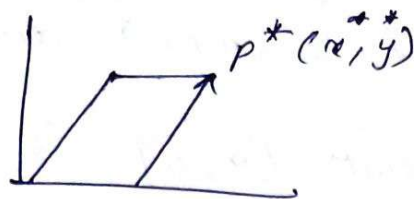
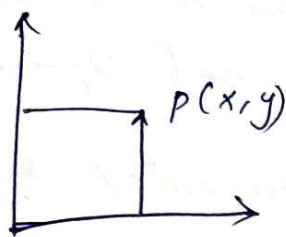
$$[P^*] = [T_m] [P]$$

for uniform

for uniform scaling $s_x = s_y$

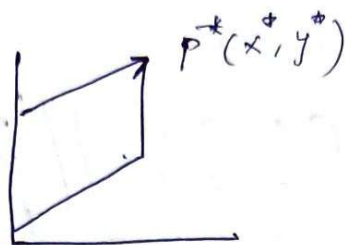
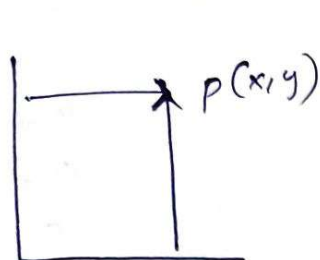
For Non Uniform scaling $s_x \neq s_y$.

shear in x-direction



$$\begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} 1 + s \sin \alpha \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$[P^*] = [S_{xy}] [P]$$



$$\begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ s \sin \alpha & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$P^* = [S_{ky}] [P]$$

Composite / Concatenated transformation.

(more than 1 transformation)

$$P^* = [S] [R] [P]$$

$[S]$ = Scaling matrix

$[R]$ = Rotation Matrix

This is a combination of 2 or more transformation such as translation, rotation, scaling etc. it will improve the performance of the graphic package.

homogeneous transformation

The various rigid body transformation discussed so far represent ~~translation~~, rotation, scaling, mirroring and shear are in the form of matrix multiplication, but translation takes in the form of vector addition. This makes inconvenient to concatenation. it is desirable to express all geometrical transformation in the form of matrix multiplication only, therefore homogeneous transformation offers to represent both translation & rotation in a unified framework.

Along Homogeneous Coordinates.

Note 1: In homogeneous representation a dimensional space is mapped to $(n+1)$ dimensional space.

Note 2: 2D point $[x, y]$ is represented by 3D as $[x, y, 1]$.

Note 3:

$$P^* \begin{bmatrix} x^* \\ y^* \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & dx \\ 0 & 1 & dy \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$T_m = \begin{bmatrix} 1 & 0 & dx \\ 0 & 1 & dy \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} x+0+dx \\ 0+y+dy \\ 0+0+1 \end{bmatrix}$$

$$= \begin{bmatrix} x+dx \\ y+dy \\ 1 \end{bmatrix}$$

Same as
translation
matrix

$$T_s = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_r = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3D transformation

Translation $\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$

Scaling $\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$

Reflection $\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} \pm 1 & 0 & 0 & 0 \\ 0 & \pm 1 & 0 & 0 \\ 0 & 0 & \pm 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$

Rotation about z-axis (xy plane)

$$\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Rotation about x-axis (yz plane)

$$\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

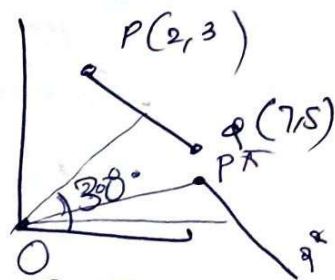
Rotation about y-axis (zx plane)

$$\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Q A line PQ with end points $P(2,3)$ & $Q(7,5)$ is to rotate about origin by 30° in clock wise direction.

Determine the coordinates of end point for rotated line

Since clock wise -ve



$$\begin{bmatrix} P^* \\ y^* \end{bmatrix} = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} \cos(-30) & -\sin(-30) \\ \sin(-30) & \cos(-30) \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{3} \times 2 + 1 \times 3}{2} \\ -\frac{1 \times 2 + \sqrt{3} \times 3}{2} \end{bmatrix} = \begin{bmatrix} \frac{3 + \sqrt{3}}{2} \\ -1 + \frac{3\sqrt{3}}{2} \end{bmatrix}$$

$$[Q^*] \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} \cos(-30) & -\sin(-30) \\ \sin(-30) & \cos(-30) \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

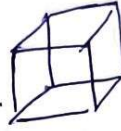
Geometric modelling

$$\begin{bmatrix} \frac{7\sqrt{3}}{2} + 4 \\ -\frac{7}{2} + 4\sqrt{3} \end{bmatrix}$$

↓

① Wire frame modelling.

- faces can't be recognised.
- Mass / Volume can't be known.



→ Geometric dimension

Topology
→ Giving relation of each parts

(2) Solid modelling.

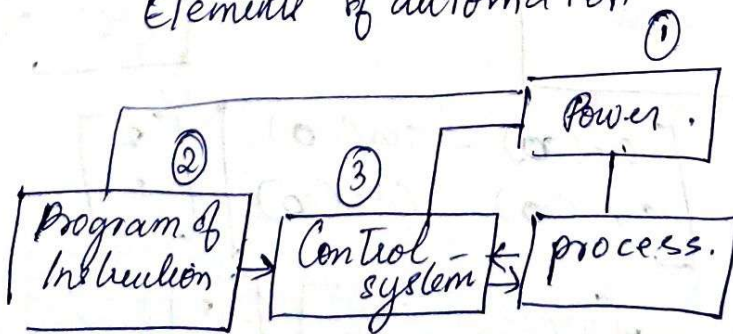
- Boundary representation method.
- Constructive solid geometry.

(3) Surface Modelling.

Automation

Use of technology to perform task with reduced human assistance.

Elements of automation

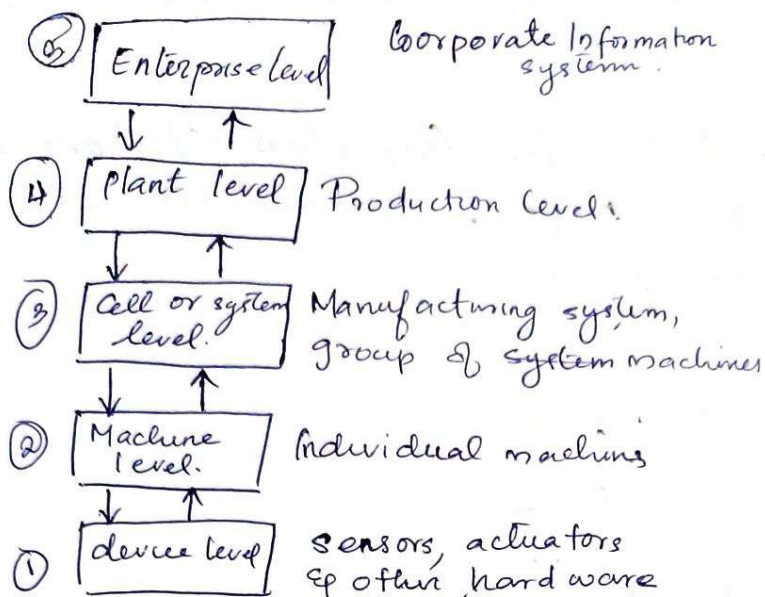


• POI - Program of Instruction

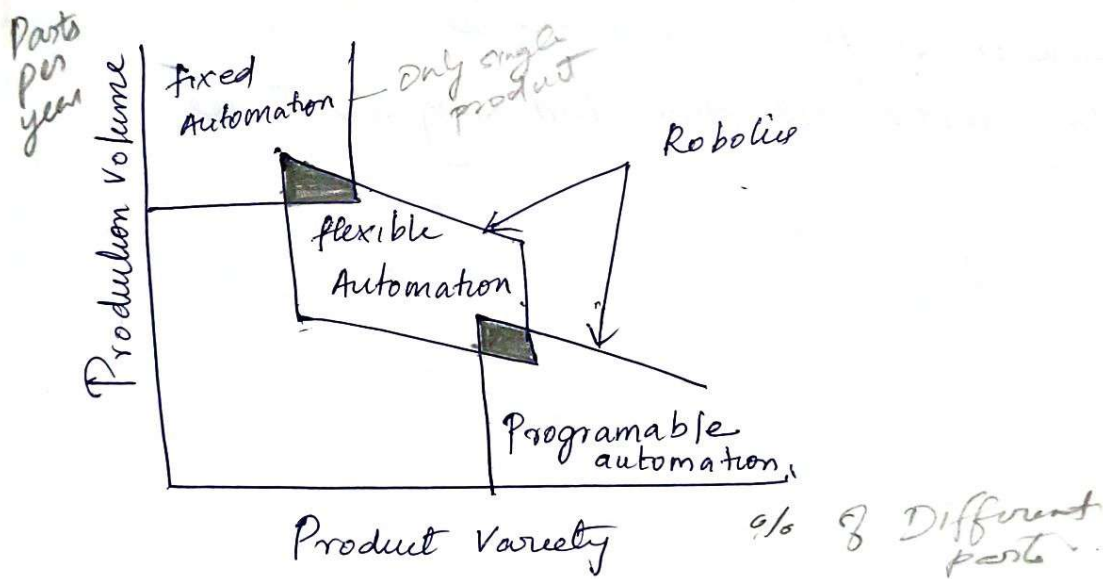
Control system

Input → Control → Actuator → Process → Output

levels of Automations



3 basic Types of Automation



7/2/24

Automated flow lines.

- The automated flow lines consist of several machines or workstation which are linked together with a work handling device, that transfer parts b/w the stations.
- A raw work - part enters one end of the line & the processing steps are performed sequentially, as the part moves from one station to another.

- it is possible to incorporate buffer storage zones in the flow line either at a single location or b/w work stations.
- It is also possible to include inspection stations in the line to automatically perform intermediate checks on the quality of work parts.

- Inline Configuration. 
- Segmented In line. 
- Rotary type configuration. 

Transfer Mechanism / work handling methods.

- linear travel for inline machines
- rotary motion for dial indexing machines.

12/2 Minimum Rational work element

Line Balancing

The main objective is to distribute task evenly over work stations so that idle time of man and machine is minimised. Line balancing aims at grouping the facilities and workers in an efficient pattern in order to obtain an optimal balance of the Capacity

2 Imp Concepts :

- Separation of total work content into minimum rational work elements.
- Precedence work Constraint must be satisfied



~~Two~~ : Summation of

$$(1) T_{wc} = \sum T_e$$

T_{wc} = Total work content time

$\sum T_e$ = Sum of work elements

$$(2) R_p = \frac{D_a}{50 \text{ Sw Hsh.}}$$

R_p = Avg hourly production rate (unit/yr)

D_a = Annual demand (Units/yr)

50 or 52 weeks/yr

Sw = no of shift/week

Hsh = Hour/shift.

$$(3) T_c = \frac{60E}{R_p}$$

T_c : Cycle time (min/cycle)

E : Uptime efficiency.

$$(4) W^* = \frac{T_{wc}}{T_c}$$

W^* : Theoretical minimum no. of workers.

$$(5) T_s = T_c - T_r$$

T_s : station time or service time

T_r : Repositioning time

$$(6) E_b = \frac{T_{wc}}{W T_s}$$

E_b = Balance efficiency.

W = Actual no. of workers.

$$(7) d = \frac{W T_s - T_{wc}}{W T_s}$$

d = Balance Delay

Problem In does

① Given $T_{wc} = 4 \text{ hr}$

$$D_a = 100000 \text{ units/yr}$$

$$\epsilon = 96/100 = 0.96$$

$$T_r = 0.08$$

$$R_p = \frac{100000}{50(5)(7.5)} = \underline{\underline{53.33 \text{ units/hr}}}$$

$$T_c = \frac{60E}{R_p} = \frac{60 \times 0.96}{53.33} = 1.08 \text{ min}$$

$$n^* = \frac{T_{wc}}{T_c} = \frac{4}{1.08} = \underline{\underline{3.7}} \approx 4 \text{ workers}$$

$$T_s = T_c - T_r = 1.08 - 0.08 = \underline{\underline{1 \text{ min}}}$$

$$E_b = \frac{D_a}{n^*} = \frac{100000}{4} = 25000 \text{ units/hr}$$

Precedence diagram: Is a graphical representation of sequence of work elements defined by precedence constraints. It is having nodes to symbolise work elements with arrows connecting nodes to indicate the order in which element must be performed.

Largest Candidate rule

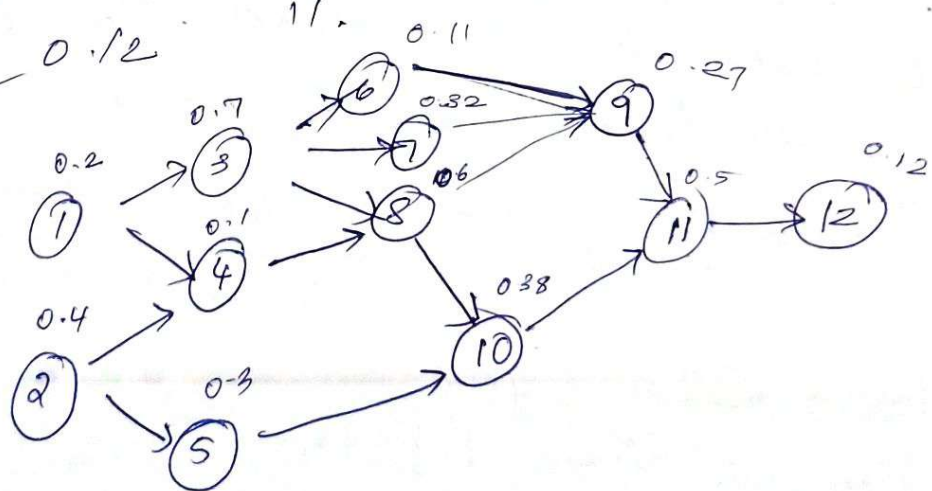
- Step 1 Draw the precedence diagram
- 2 Work elements are arranged in a descending order according to their (T_e) values.

3. Assign elements to the first work station, and by starting at the top of the list, and selecting the first element that satisfy precedence requirements and does not cause the total sum of T_e at that station to exceed allowable station time t_c

①

element No	T_e	Must be Preceded by
1	0.2	—
2	0.4	—
3	0.7	1
4	0.1	1, 2
5	0.3	2
6	0.11	3
7	0.32	3
8	0.6	3, 4
9	0.27	6, 7, 8
10	0.38	5, 8
11	0.5	1, 10
12	0.12	11

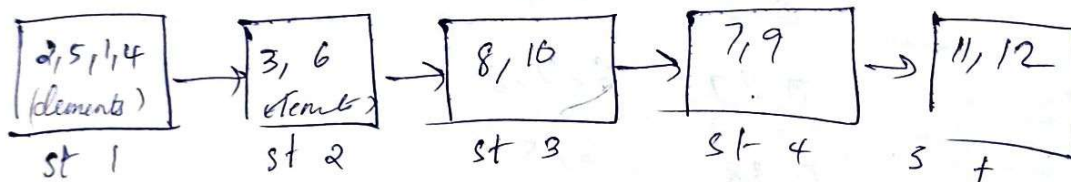
Consider
Station time max 1
No of worker in each station 1
→ Calculate work stations & no of workers required
→ find balance efficiency and delay



Precedence Diagram.

work element	t_e	precedent
3	0.7	1
8	0.6	3, 4
11	0.5	9, 10
2	0.4	-
10	0.38	5, 8
7	0.32	3
5	0.3	2
9	0.27	6, 7, 8
1	0.2	-
12	0.12	11
6	0.11	3
4	0.1	1, 2

Work flow



Since the system having 5 work station with each station having 1 worker; Therefore we need 5 workers to complete the process.

$$\text{Balance efficiency } E_b = \frac{\text{Total work content time}}{\text{no. of worker} \times \text{station time}} \quad \frac{T_{wc}}{W \times T_s}$$

$$T_{wc} = \sum T_e = 4 \text{ min} \quad = \frac{4}{5} = 0.8$$

Here we have 5 stations = $W = 5 \text{ N} = 5 \text{ worker}$

$$T_s = 1 \text{ min}$$

$$T_e = 0.8 \text{ min}$$

1 worker / station

step 3			
station	Work element	T_e	
1	2 5 1 4	0.4 0.3 0.2 0.1	1 min
2	3 6	0.7 0.11	0.8
3	8 10	0.6 0.38	0.9
4	7 9	0.32 0.27	0.59
5	11 12	0.5 0.12	

Allowable station time = 1 min

element no	T_e preceded by
1	0.2
2	0.4
3	0.7
4	0.1
5	0.3
6	0.11
7	0.32
8	0.6
9	0.37
10	0.1
11	
12	

$$\text{Balance delay} = d = \frac{WT_s - T_{wc}}{WT_s}$$

$$= \frac{5 \times 1 - 4}{5 \times 1}$$

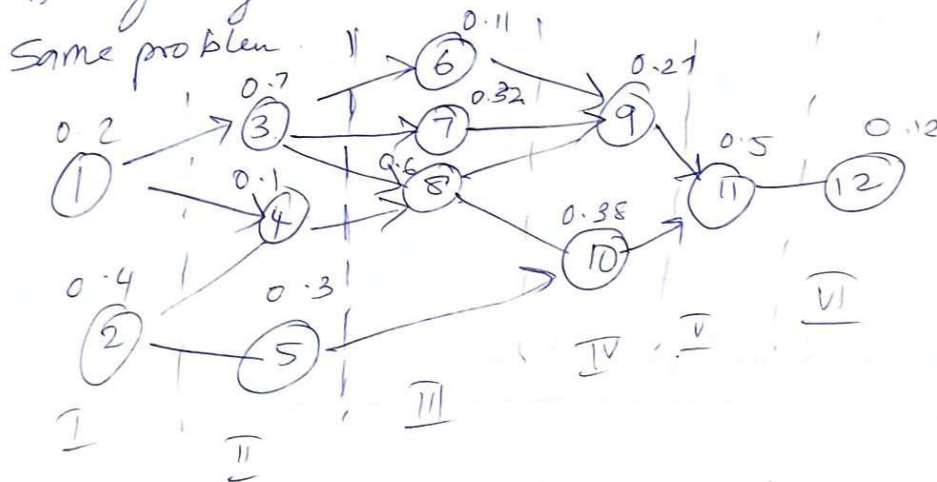
$$= 0.2 = 20\%$$

Killbridge and western method (KWTT)

1) constructing the precedence diagram in terms of columns.

2) Arranging the work elements according to columns.

3) Assigning work elements to the station.



work elements	column	$T_{ec}(\text{mean})$	station time
1	I	0.2	0.6
2	II	0.4	
3	III	0.7	1.1
4	IV	0.1	
5	V	0.3	
6	VI	0.11	1.3
7	VII	0.32	
8	VIII	0.6	

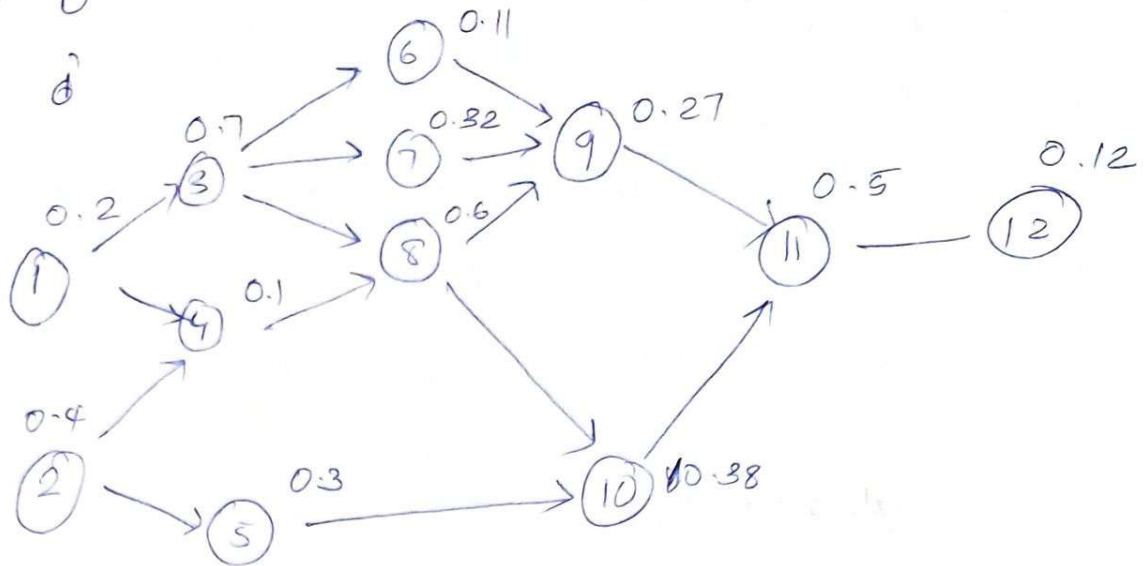
9	<u>TV</u>	0.27	0.65
10		0.38	
11	<u>V</u>	0.5	0.6
12	VI	0.12	0.12

station	Element	T_e	station Σt_e
<u>I</u>	1	0.2	1.0
	2	0.4	
	4	0.1	
	5	0.3	
<u>II</u>	3	0.7	0.81
	6	0.11	
<u>III</u>	7	0.22	0.92
	8	0.6	
<u>IV</u>	9	0.27	0.65
	10	0.38	
<u>V</u>	11	0.5	0.62
	12	0.12	

Rank positional weight method.

- 1) Draw the precedence diagram.
- 2) Calculate RPW value for each element.
for eg. Consider element 1, the elements that follow in the arrow chain are 3, 4, 6, 7, 8, 9, 10, 11 and 12. The rpw value for element 1 would be the sum of element time T_e for all these elements.
- 3) Arrange the work element according to descending order of RPW

Assigning work elements to the station.



$$(RPW)_1 = 0.2 + 0.7 + 0.11 + 0.27 + 0.5 + 0.12 + 0.32 + 0.6 + 0.38 + 0.1$$

$$= \underline{\underline{3.3}}$$

$$(RPW)_2 = 0.4 + 0.1 + 0.6 + 0.27 + 0.5 + 0.12 + 0.3 + 0.38$$

$$= \underline{\underline{2.67}}$$

$$(RPW)_3 = 0.11 + 0.32 + 0.6 + 0.27 + 0.5 + 0.12$$

$$= \underline{\underline{2.97}}$$

$$(RPW)_4 = 0.1 + 0.6 + 0.27 + 0.5 + 0.12 + 0.38$$

$$= 1.97$$

$$(RPW)_5 = 0.3 + 0.38 + 0.5 + 0.12 = 1.3$$

$$(RPW)_6 = 0.11 + 0.27 + 0.5 + 0.12 = 1.0$$

$$(RPW)_7 = 0.32 + 0.27 + 0.5 + 0.12 = 1.21$$

$$(RPW)_8 = 0.6 + 0.27 + 0.5 + 0.12 = 1.49$$

$$(RPW)_9 = 0.27 + 0.5 + 0.12 = 0.89$$

$$(RPW)_{10} = 0.38 + 0.5 + 0.12 = 1.0$$

$$(RPW)_{11} = 0.62$$

$$(RPW)_{12} = 0$$

work element	RPW	Te	Precedence.
1	3.3	0.2	—
3	3	0.7	1
2	2.67	0.4	—
4	1.97	0.1	1, 2
8	1.87	0.6	3, 4
5	1.3	0.38	2
7	1.21	0.32	3
6	1	0.11	3
10	1	0.38	5, 8
9	0.89	0.27	6, 7, 8
11	0.62	0.5	9, 10
12	0.12	0.12	11

Station	element	Te	station time
<u>I</u>	1	0.2	0.9
	3	0.7	
<u>II</u>	2	0.4	0.91
	4	0.1	
	5	0.3	
	6	0.11	
<u>III</u>	8	0.6	0.92
	7	0.32	
<u>IV</u>	10	0.38	0.65
	9	0.27	
<u>V</u>	11	0.5	0.62
	12	0.12	

Note : In RPW method, Balanced efficiency and Balance delay are calculating by considering the station time T_s as a maximum station process time

In the above problem it is clear that max station time equal to 0.92 mins.

$$\eta = \frac{\cancel{N T_s} - T_{wc}}{N T_s}$$

$$T_{wc} = \epsilon T_e$$

$$\text{Balance delay} = \frac{N T_s - T_{wc}}{N T_s}$$

$$= \frac{5 \times 0.92 - 4}{5 \times 0.92} = \underline{\underline{0.130}}$$

$$\eta = \frac{T_{wc}}{N T_s} = \frac{4}{5 \times 0.92} = 0.869 = 86.9\%$$

Nc words

Sequence words (N)

Preparatory words (G)

(Check photo in MEA for G & M codes)

Coordinate (X, Y, Z)

Feed rate (F words)

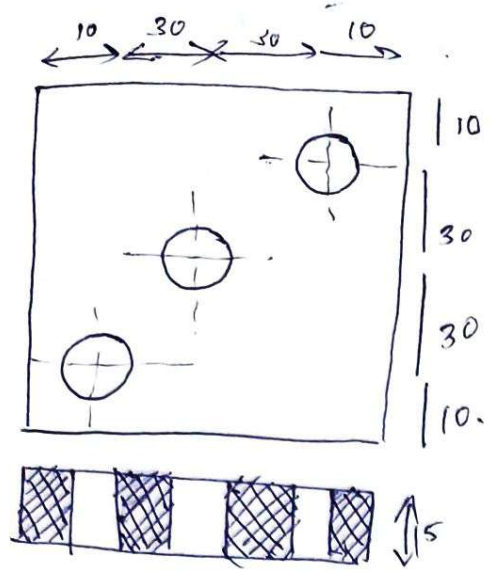
Cutting speed (S)

Tool selection (T words)

Miscellaneous ~~no~~ functions (M)

18 + 8 + 14 + 12

UNIT 10



G81 - Drilling cycle
 G80 - Drilling cycle ~~cancel~~
 G99 - Repeat Point return

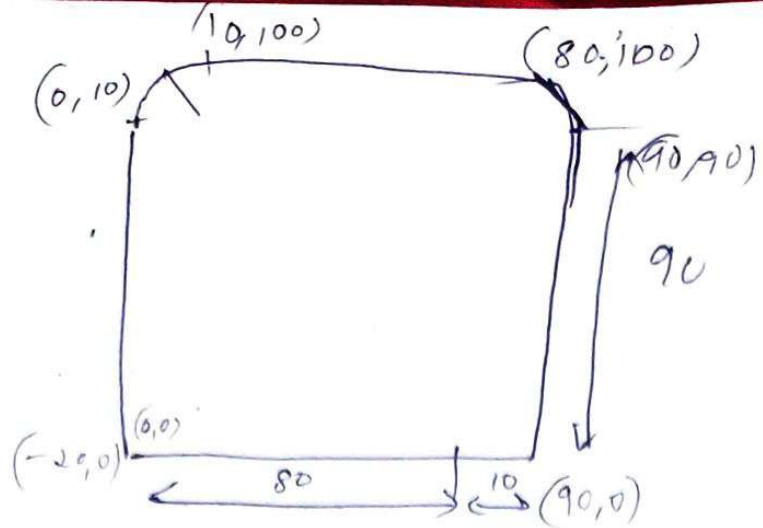
G81 - X - Y - Z - R - F

G Code



- N01 G00 - G28
- N02 M06 T03
- N03 G90 G00 X10 Y10 Z10 S1100 M03 M08
- N04 G99 G81 X10 Y10 Z17 R2 F100
- N05 X48 Y40
- N06 X70 Y70
- N07 G00 G80 Z10
- N08 G00 M05 M09
- N09 G00 G28
- N10 M30

102



~~N01 G21 G90 G94~~
~~N02 G00 X20 Y0~~
~~N03 G00 Z10~~
~~N04 G01 G42 D20~~

N01 G21 G90 G94
 N02 G00 X20 Y0
 N03 G00 Z10
 N04 G01 G42 D20 Z10 S1000 F200 M03
 N05 G01 X0 Y0
 N06 G01 X90
 N07 G01 Y90
 N08 G01 X80 Y100
 N09 G01 X10
 N10 G03 X0 X90 R10
 N11 G01 Y0
 N12 G40
 N13 G00 X20 Y0